

TLE Differencing

Post Hoc Uncertainty Estimation

A.C.

May 26, 2026

Outline

TLE Primer

- Kepler Theory

- Mean Element Theories

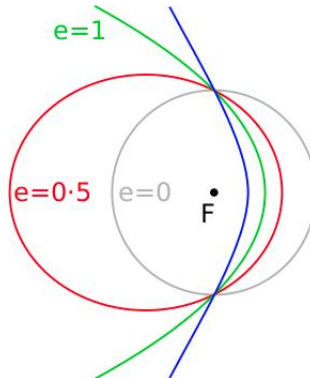
- Limitations

TLE Differencing

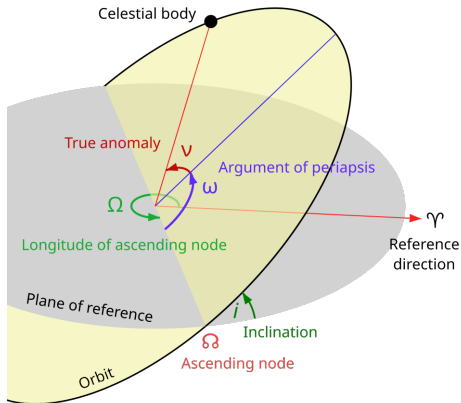
Conic Perfection

$$\ddot{\mathbf{R}} = \frac{\mu}{r^2}$$

Conic Perfection

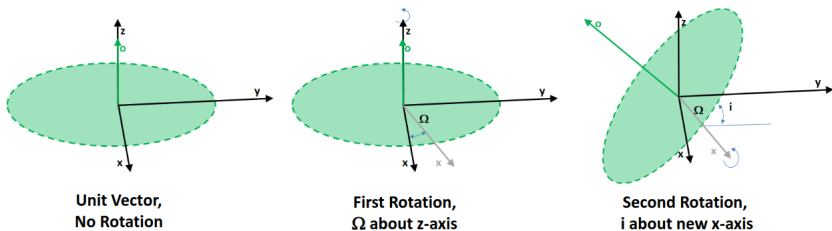


Kepler Elements



Element	Function
i	Orientation
Ω	
ω	
e	Shape
a	
ν	Position

Kepler and Euler



Kepler elements result in busy diagrams, but can be quite handy.
E.g., i and Ω directly define the direction of orbital momentum.

Perturbation: Non-Sphericity

$$\ddot{\mathbf{R}} = \ddot{\mathbf{R}}_{2B} + \ddot{\mathbf{R}}_{NS}$$

- ▶ $\ddot{\mathbf{R}}_{2B}$ uses point-mass equations
 - ▶ Works for points
 - ▶ Works for spheres (shell theorem)
- ▶ Earth is neither
 - ▶ Centrifugal forces (order kilometers)
 - ▶ Tidal forces (order meters)
- ▶ Notable for causing nodal precession.

Perturbation: Drag

$$\ddot{\mathbf{R}} = \ddot{\mathbf{R}}_{2B} + \ddot{\mathbf{R}}_{NS} + \ddot{\mathbf{R}}_D$$

- ▶ Drag equation: $F_D = \frac{1}{2}\rho v^2 C_D A$
- ▶ Scales by
 - ▶ Object shape and orientation (C_D, A).
 - ▶ Square of object velocity v^2 .
 - ▶ Atmospheric density ρ , drops exponentially with altitude.
- ▶ LEO objects (altitude < 2000 km) are low and fast
 - ▶ experience non-negligible drag
- ▶ Bonus: non-periodic and non-conservative.

Perturbation: et cetera

We've only scratched the surface.

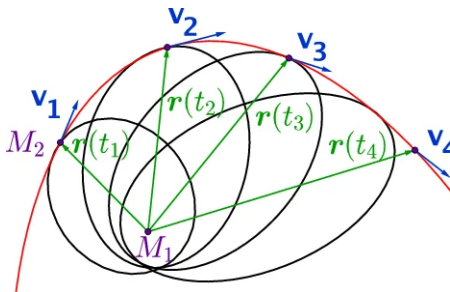
- ▶ Third bodies (esp. sun, moon)
- ▶ Tidal forces (solid and not)
- ▶ Solar radiation pressure (esp. HAMR)
- ▶ Knocke rediffusion
- ▶ Relativistic corrections
 - ▶ Schwarzschild
 - ▶ Lense-Thirring
 - ▶ De Sitter

Osculating Elements

Perturbed orbits will *not* be ellipses. Can we pretend?

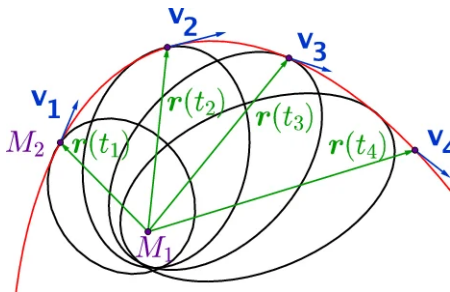
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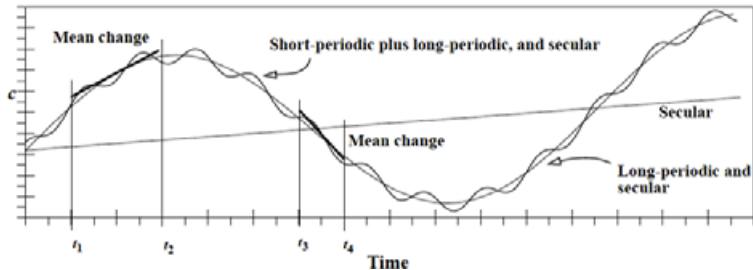
Osculating Elements

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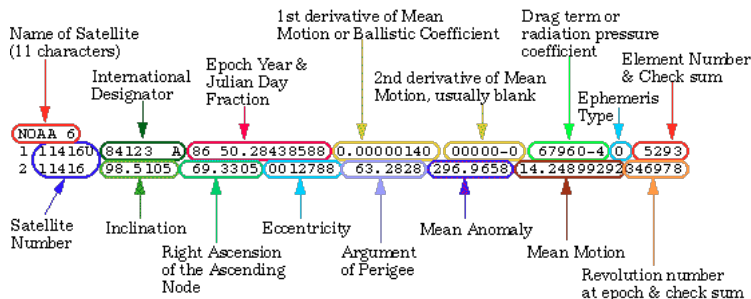
Yes, but...

Mean Elements



Could we “average” the high-frequency perturbations away somehow?

SGP4 and TLEs



Propagation Error

$$\ddot{R}_{\text{SGP4}} = \ddot{R}_{2\text{B}} + \ddot{R}_{\text{NS}} + \ddot{R}_{\text{D}} + \ddot{R}_{\text{sun}} + \ddot{R}_{\text{moon}}$$

- ▶ SGP4 catches only the biggest perturbations.
- ▶ Liable to accumulate kilometers of error within an orbital period.



Error at Epoch

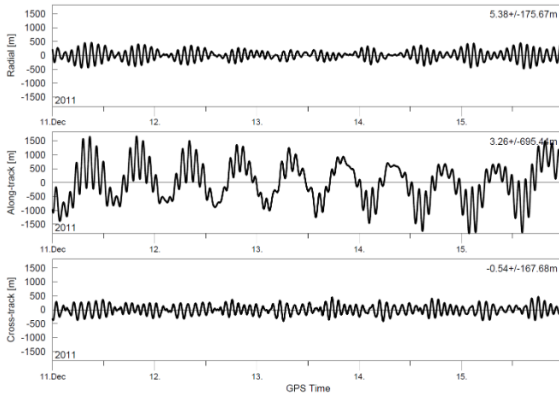


Figure 1 Orbit Error of the Fitted TLE during OD Period (5d Fit)

Precision

All models are wrong. Can we get estimates on uncertainty?
Perhaps a covariance matrix?

Precision

404

Outline

TLE Primer

TLE Differencing

General Approach

Subtleties

Experimental Verification

Differencing

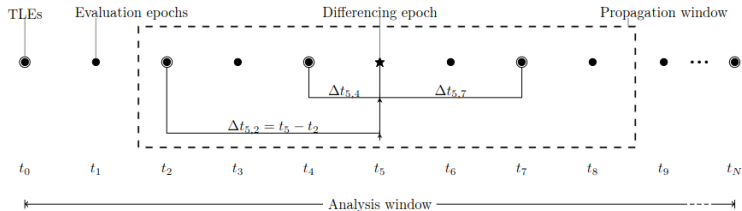


Figure 1: Schematic overview of TLE differencing method.

1. Select an RSO's TLE history to examine.

Differencing

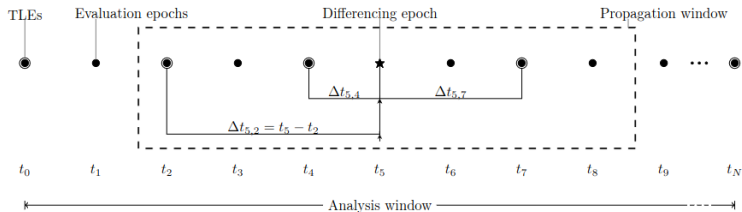


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2. Assume residuals ϵ are consistently distributed over an *Analysis Window*.

Differencing

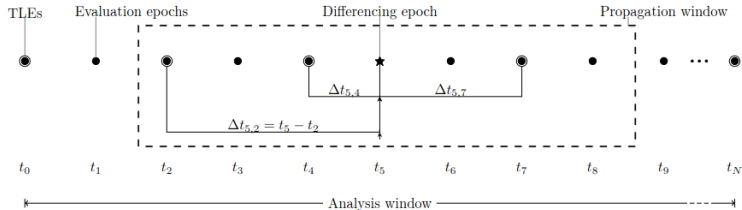


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3. Choose a *Propagation Window* over which SGP4 should be “good enough”.

Differencing

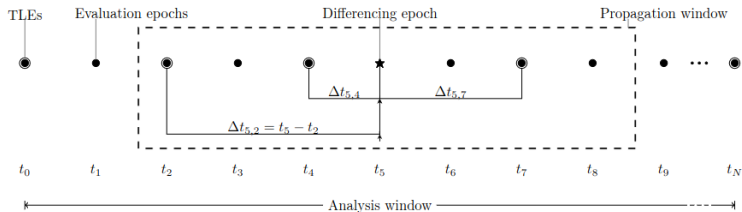


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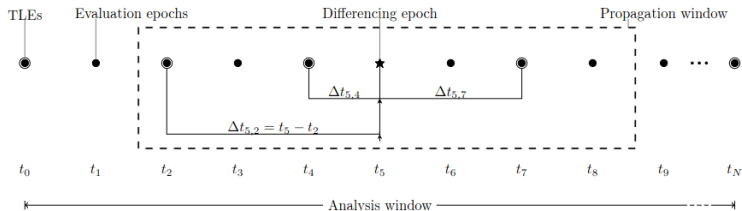


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Differencing Recapitulated

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Fit to Variance

Recall: standard OLS model

$$y = X\beta + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

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Prediction error spread:

$$\text{Var}(y_{\text{new}} - x_{\text{new}}\beta) = \sigma^2 + \sigma^2 x_{\text{new}}(X^T X)^{-1} x_{\text{new}}^T$$

Example Results

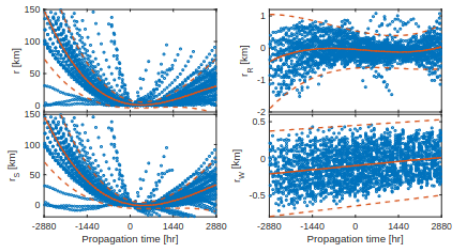
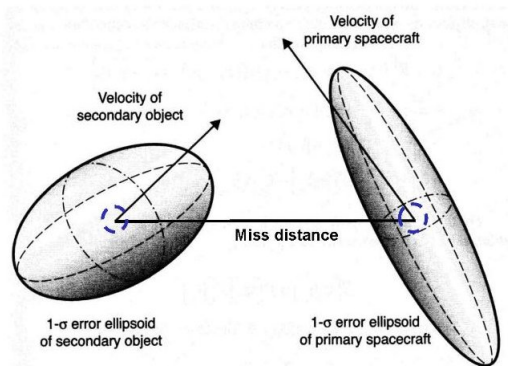


Figure 4: Robust regression of the position errors and (3σ) uncertainty bounds decomposed in the RSW frame.

Reference Frame



Residuals naturally oriented along satellite motion (RIC frame).

Reference State

Using TLEs at epoch as reference states limits our pool of residual samples. What if we merged all our TLEs together into a reference state, a la sample means?

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Why inverse square? No theoretical justification provided.

Propagation Asymmetry

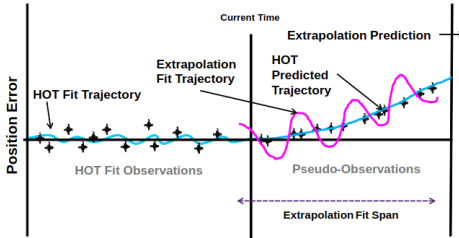
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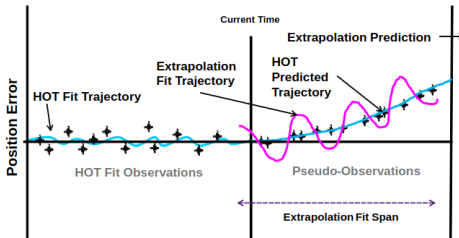


Figure 1: Extrapolation DC fitting approach diagram.

In fact TLEs may not be most accurate at epoch!

Error Model

Recall: final step of differencing protocol was to fit an error estimating function $\hat{\epsilon}(\Delta t)$.

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Polynomials? Rationals? ANNs?

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Just interpolating, data is “well-behaved” – low order polynomial is fine. Plus, interacts well with OLS.

Outlier Handling: Rejection Multipliers

TLEs are crap sometimes – need to manage outliers.

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Commonly effective, but $r_{\max}$ is a nuisance hyper-parameter.

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WLSQ and Bi-square weight function are topics for another day, but in summary:

1. WLSQ minimizes a weighted L2 norm.
2. Bi-square weight gives more weight to well-modeled obs and less to poorly modeled obs.

GNSS

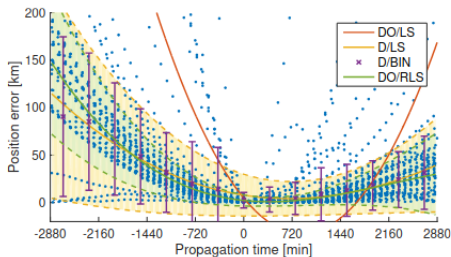


Figure 3: Benchmark of different methods for estimating the error and (3σ) uncertainty of the TLE propagation errors of Period 1a.

GNSS readings (blue) provide truth baseline for evaluation.

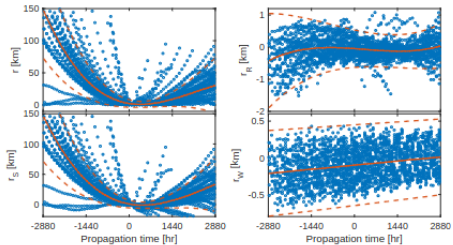


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As expected, in-track error dominates.

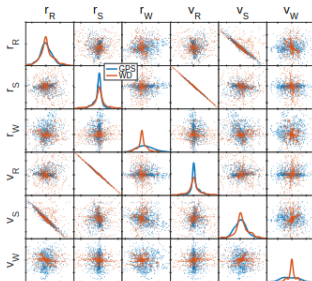


Figure 9: Correlation estimation using weighted differences compared to GPS of classic TLEs.

Independent polyfits do not covariance predict. But combined with correlations from raw sample statistics...

References and Further Reading

- [1] Béatrice Deguine et al. "Covariance modelling in satellite collision risk activities". In: *AIAA/AAS Astrodynamics Specialist Conference and Exhibit*. 2002, p. 4631.
- [2] Saika Aida and Michael Kirschner. "Accuracy assessment of SGP4 orbit information conversion into osculating elements". In: (2013).
- [3] MD Hejduk et al. "A catalogue-wide implementation of general perturbations orbit determination extrapolated from higher order orbital theory solutions". In: *Proceedings of the 23rd AAS/AIAA space flight mechanics meeting*. 2013, pp. 619–632.
- [4] Jacco Geul, Erwin Mooij, and Ron Noomen. "TLE uncertainty estimation using robust weighted differencing". In: *Advances in Space Research* 59.10 (2017), pp. 2522–2535.

Self Link

The source for this presentation is hosted at <https://github.com/alan-christopher/edu/tle-differencing>.